

Giri's Tech Hub, Pune.

Core Java Machine Test

Nov-24 to June-25

Time - 08 to 11 Am

Instructions:

1. Solve any 10 questions.
2. Input should be from user.
3. Indentation and comments mandatory.
4. Without using any inbuilt functions.

Q1. Write a java program to print this pattern.

```
      1
     2 2
    3 3 3
   4 4 4 4
  5 4 3 2 1 2 3 4 5
  4 3 2 1 2 3 4
   3 2 1 2 3
    2 1 2
     1
```

Q2. Design a queue using an array. Remove all occurrences of a given element.

Explanation

Traverse and skip matching elements.

Input - Queue: 10, 20, 10, 30

Remove: 10

Output - Queue: 20 30

Q3. Create a class named PrimeArrayManager using a parameterized constructor that accepts an integer array.

The constructor should initialize the array, and the class should provide methods to perform the following operations:

1. Separate all Prime Numbers and Non-Prime Numbers.
2. Sort the Prime Numbers in Ascending Order.
3. Sort the Non-Prime Numbers in Descending Order.
4. Merge both arrays such that all sorted prime numbers appear first, followed by all sorted non-prime numbers.
5. Display the Prime Numbers, Non-Prime Numbers, and the Final Merged Array. Do not use Java Collections or built-in sorting methods. Implement the sorting logic manually.

Input - Enter the size of array - 8

Enter the array elements - 15 7 11 20 13 4 17 9

Output - Prime Numbers : 7 11 13 17

Non Prime Numbers : 20 15 9 4

Final Array : 7 11 13 17 20 15 9 4

Q4. Given N strings, group together all strings having the same characters irrespective of order.

Do not use collections or sorting methods.

Explanation - listen silent enlist belong to same group.

**Input - Enter number of strings: 6
listen silent abc cab enlist dog**

Output - Group 1	Group 2	Group 3
listen	abc	dog
enlist	cab	
silent		

Q5. Write a Java program to create a banking application to withdraw money with exception handling.

Description: Accept balance and withdraw amount. If the withdrawal amount is greater than the balance, throw a custom exception.

Input:

Enter balance: 5000

Enter withdraw amount: 7000

Output: Insufficient Balance Exception

Q6. Add two matrices using multiple threads (one thread per row).

Example Input:

Matrix A = {{1,2},{3,4}}

Matrix B = {{5,6},{7,8}}

Example Output:

Result = {{6,8},{10,12}}

Explanation:

Each thread calculates one row of result matrix. Synchronization not needed if each thread works on separate row.

Q7. Store N integers in an ArrayList. Rotate the elements to the right by one position.

Input:

5

10 20 30 40 50

Output:

Rotated List: 50 10 20 30 40

Description:

Store elements in ArrayList.

Move last element to first position

manually. Shift remaining elements one position forward. Do not use

Collections.rotate().

Q8. Store N words in a HashMap. Print words whose length is a prime number.

Input: java python sql code spring

Output: Words with Prime Length :

java

python

Description:

- Store words in a HashMap.
- Calculate the length of each word.
- Check whether the length is a prime number.
- Print all matching words.

Q9. Store N words in a HashMap. Print the words that contain at least one consecutive repeated character.

Input: apple book java hello moon

Output: Words with Consecutive Repeated Characters:

apple

book

hello

moon

Description:

- Store words in a HashMap.
- Traverse each word character by character.
- Print words containing consecutive repeated characters (e.g., "pp", "oo", "ll").

Q10. Store N Words in a HashMap and Print Words with Unique Character Frequencies.

Write a Java program to store N words in a HashMap. For each word, calculate the frequency of every character. Print only those words in which the frequency of every distinct character is unique.

If two or more characters have the same frequency in a word, that word should not be printed.

Input

6

aabbccc

aabbcc

banana

abcde

mississippi

programming

Output

Words with Unique Character Frequencies:

aabbccc

banana

abcde